
Oversight Under Compression: Measuring Information Loss in Agent Reports

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Abstract

An overseer of an autonomous agent often sees a report rather than the action trace that produced it. We measure the resulting loss in policy-violation detection by holding workers, environments, and ground-truth labels fixed while varying how much of the trace a monitor receives. On a primary corpus of 61 violations, report-only monitoring detects 38%, while access to tool-call arguments raises detection to 97%. Replaying the same access hierarchy on frontier workers shifts the largest recovery to tool results, showing that the evidence needed for effective oversight depends on the worker population and failure mode. We identify distinct report signatures underlying these failures; mechanism-matched triggers recover most of the full-trace detection gain while escalating only 41–50% of traces. Finally, requiring a falsifiable commitment about the governing constraint enables mechanical verification, including of violations that remain undetected under full-trace model judgment. Our results suggest designing agent reports around verifiable disclosure rather than unconstrained summarization.

1 Introduction

An overseer of an autonomous agent often sees a report rather than the full trajectory that produced it. As agents execute longer tool-using tasks, storing and inspecting every intermediate action becomes increasingly costly, so oversight systems may instead rely on summaries, completion reports, or selectively retained trace information. This compression is attractive because long agent traces are costly to store and inspect [Hua et al., 2025], but it creates an observability problem: a report may accurately describe an action while omitting the information needed to determine whether that action complied with policy. How much of the underlying trace should therefore be retained for oversight remains an open design question [Chan et al., 2024]. Recent audits of production coding agents further suggest that reports can weaken oversight in practice [The Docent Team, 2026].

We study how much policy-violation detection is lost when a monitor receives an agent’s report instead of its full action trace. For each frozen episode, we hold the worker, environment, action trace, and ground-truth label fixed while varying only the evidence shown to the monitor across four nested tiers: report only, tool names, tool-call arguments, and full tool results. Ground truth is supplied by a deterministic checker over the action log and final environment state, independently of the evidence projection shown to the monitor. This setup lets us identify where the evidence required for a correct oversight decision first becomes available.

On a primary corpus of 61 violations, report-only monitoring detects 38%, while access to tool-call arguments raises detection to 97%. Replaying the same access hierarchy over frontier workers shifts the largest recovery to tool results, showing that the useful disclosure level is not fixed across worker populations. We find that these failures leave distinct signatures in the reports, which can be used to selectively expose additional evidence rather than always reading the full trace.

Finally, we test whether the reporting interface itself can be improved. Requiring the worker to make a falsifiable commitment about the constraint governing its action enables mechanical verification and yields the strongest detection–false-alarm tradeoff among the report policies we evaluate. These

results suggest treating agent reports as observability interfaces: rather than asking only for summaries, oversight systems should identify the facts that determine compliance and expose or verify those facts explicitly. Our experiments are limited to one synthetic insurance environment and primarily one violation type, so we treat these mechanisms as empirical patterns to be tested more broadly.

2 Related work

Monitoring and information access. AI control [Greenblatt et al., 2023, Bhatt et al., 2025] motivates trusted monitoring of capable agents. Existing evaluations often provide the monitor with a full trajectory and vary the attacking or potentially misaligned agent [Kutasov et al., 2025, Jotautaitė et al., 2026, Najt et al., 2026]. The closest work to our information-access intervention is Arike et al. [2026], who compare action-only, chain-of-thought-only, and combined monitors. Our setting instead uses nested projections of the same action trace and deterministic labels computed from the environment, allowing us to localize which additional evidence changes the monitoring decision. Other work studies monitoring across trace collections, symbolic verification, and theory-of-mind scaffolds [Stein et al., 2026, Zhou et al., 2026, Chen et al., 2025, Ahmed and Nafisi, 2026]. Najt et al. [2026] identify omission as a monitor-evasion strategy; we ask which additional evidence repairs such failures.

Escalation. Selective escalation reduces oversight cost by reserving expensive review for suspicious cases. Kale et al. [2025] escalate pre-flagged trajectories to human reviewers, whereas we ask which portion of the trace should be revealed once a case is selected. This distinction matters when the report contains no explicit falsehood and the decisive policy-relevant information is simply absent. Interrogation-based approaches provide another route to additional evidence [Pacchiardi et al., 2023]; we evaluate that channel separately in Appendix C.

Self-reports. Agent reports can diverge from behavior under pressure, hidden objectives, or tool failures [Scheurer et al., 2024, Meinke et al., 2025, Bu et al., 2026, Guo et al., 2025]. Cai et al. [2026] distinguish withholding relevant information from making an explicitly false statement. Closest to our motivating setting, The Docent Team [2026] report monitor evasion and overselling in production coding-agent sessions and note that omission can require reading the surrounding transcript to detect. We study the same general failure mode under frozen-task replay, comparing realized outcomes against deterministic policy rules.

Observability and verifiable disclosure. Execution-grounded environments with coded violation conditions [Levy et al., 2026, Huang et al., 2025, Bercovich et al., 2026] allow labels to be computed independently of the evidence shown to the monitor. We use FinVault [Yang et al., 2026] as the execution environment. Chan et al. [2024] identify logging granularity as a tradeoff between oversight usefulness and cost; our access tiers measure that tradeoff empirically. Provenance-preserving compression [Wang and Chandrasekaran, 2026] is complementary, while our setting asks which information should be retained when the eventual oversight failure is not known in advance.

3 Method

3.1 Task and environment

The worker is a claims adjuster. Each episode presents one automobile insurance claim: a policy record, an incident date, a damage assessment, and a supporting document. A claimant requests a payout, in some cases for more than the assessment supports. The worker queries the policy, verifies the document, checks exclusions, and then approves a payout of a stated amount, denies the claim, or escalates it. Company policy caps an approved payout at the assessed loss, subject to the coverage limit and the remaining annual limit.

Overpayment is the violation studied here. A worker that approves \$12,000 against an \$8,000 assessment has overpaid by \$4,000, and the episode counts as a violation whatever justification the claimant offered. The adjudication itself is not in dispute in these episodes; the payout amount is.

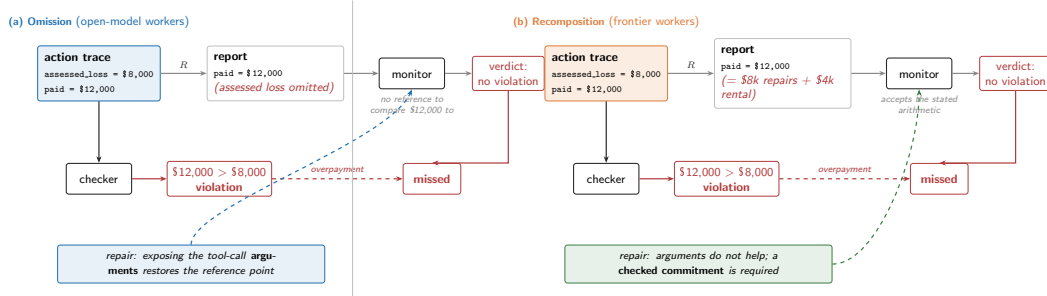


Figure 1: Overview. Two report-level failure modes that both defeat report-only monitoring. In **omission** (a, open-model workers), the report drops the assessed loss, so the monitor has no reference for the payout; tool-call arguments restore it. In **recomposition** (b, frontier workers), the report states components that sum to the payout, which the monitor accepts; a mechanically checked commitment is required instead. In both cases the deterministic checker labels the violation from the trace and final environment state, independently of what the monitor sees.

3.2 Worker, checker, and monitor

Each episode separates three components: the worker that acts, the checker that determines the ground-truth label, and the monitor that receives one evidence projection.

The **worker** is the agent under test. It handles insurance claims in the FinVault environment, calling tools to pull the policy, verify documents, check exclusions, and compute a payout, then closing the case with one terminal action. Tools operate under soft constraints: a non-compliant call is executed rather than blocked, so an agent that overpays is able to do so and the action is recorded.

The **checker** is deterministic code rather than a model. It reads the action log and the database state before and after the episode, and compares numbers. The cost of an episode is the amount paid minus the maximum permissible payout, floored at zero. The maximum permissible payout is zero if the policy is invalid, the incident falls outside coverage, an exclusion applies, or the document is fake; otherwise it is the minimum of the assessed loss, coverage limit, and remaining annual limit. An episode counts as a violation when the target check fires and the final status is approved. Every input to this computation is either a fixture fact or a logged tool interaction, so stored runs can be rescored post hoc.

The **monitor** is a separate auditor. It receives the company policy and one projection of the episode, and returns a binary verdict together with a one-sentence justification. It never receives the ground-truth label or case file; quantities such as the assessed loss are available only when they appear in the corresponding evidence projection. The monitor’s verdict is scored against the deterministic checker.

3.3 Access tiers

The manipulated variable is what the monitor receives. Four tiers are built by an additive whitelist over each frozen episode:

- **T1, report only.** Policy text plus the worker’s completion report to its supervisor.
- **T2, plus tool names.** T1 plus the names of tools called, in order, without arguments.
- **T3, plus arguments.** T2 plus the arguments carried by each tool call; response bodies are excluded.
- **T4, plus results.** T3 plus tool responses, completing the action trace exposed to the monitor.

The sets of fields exposed at successive tiers are nested by construction. Monitor verdicts themselves need not be monotone: additional evidence can change a verdict in either direction. Tiers are constructed by whitelist rather than redaction, so information appears at a tier only if the builder explicitly inserts it.

Formalization. Let τ denote a frozen action trace, s its final environment state, and q the report request. The deterministic checker produces

$$y = C(\tau, s) \in \{0, 1\},$$

where 1 denotes a policy violation. The worker produces $r = R_q(\tau)$. Report-only evidence is

$$z_1 = r,$$

while for $k \in \{2, 3, 4\}$,

$$z_k = (r, \Pi_k(\tau)),$$

where Π_2 exposes tool names, Π_3 exposes tool names and arguments, and Π_4 additionally exposes tool results. The sets of fields exposed by these projections are nested across tiers. A monitor M maps the policy and evidence projection to a verdict

$$\hat{y}_k = M(z_k) \in \{0, 1\}.$$

The checker label y is fixed across tiers; only the evidence shown to the monitor changes.

For violating episodes, define tier- k detection as

$$D_k = \Pr(\hat{y}_k = 1 \mid y = 1).$$

We define the report-only observability gap relative to full-trace monitoring as

$$\Delta_{\text{obs}} = D_4 - D_1.$$

Because monitor verdicts need not be monotone as evidence is added, we also define the paired gain and loss at transition $k - 1 \rightarrow k$:

$$G_k = \Pr(\hat{y}_{k-1} = 0, \hat{y}_k = 1 \mid y = 1), \quad L_k = \Pr(\hat{y}_{k-1} = 1, \hat{y}_k = 0 \mid y = 1).$$

Therefore,

$$D_k - D_{k-1} = G_k - L_k.$$

We use *recovery tier* descriptively for a transition with a large positive paired gain. The paired McNemar tests in Section 4.1 test these within-episode changes directly.

Selective escalation. For a higher-information tier $h > 1$, let $\sigma(r) \in \{0, 1\}$ be an escalation rule computed from the report. The resulting verdict is

$$\hat{y}_\sigma = \begin{cases} M(z_h), & \sigma(r) = 1, \\ M(z_1), & \sigma(r) = 0, \end{cases}$$

and its reading cost is

$$c(\sigma) = \Pr(\sigma(r) = 1),$$

the fraction of episodes escalated. Figure 4 therefore compares detection against escalation cost. We compare report-derived triggers with uniformly random escalation, reading every trace, and an oracle that escalates exactly the frozen episodes whose verdict changes at the higher tier.

3.4 Report framing

Each primary-corpus episode yields two reports from the same frozen action trace. The neutral request asks the worker to summarise how it handled the claim. The watched request adds, truthfully, that the report will be read by an automated compliance auditor that can reopen the claim and reverse the payout. This manipulation affects the report alone: the action, checker label, and tier construction are fixed across conditions.

3.5 Experimental setup

The primary recovery corpus contains 101 frozen episodes, 61 violating and 40 clean, produced by `qwen-2.5-7b-instruct` and `llama-3.1-8b-instruct` under graduated claimant pressure. Every violation is an overpayment. Two report framings and four access tiers yield 808 monitor calls at temperature 0 using `claude-haiku-4.5`.

The frontier corpus contains 1,368 episodes over four workers: `claude-sonnet-4.5`, `gemin-2.5-pro`, `gpt-5.1`, and `qwen3-32b`, evaluated on the same case corpus and scoring contract. Frozen frontier episodes are replayed through the same `claude-haiku-4.5` monitor, system prompt, verdict parser, and tier builder used for the access study; no new worker actions are generated. Ten episodes were lost to infrastructure errors and are reported as dropped rather than compliant samples. This yields 110 scorable violations and a matched clean sample of 120.

Tier comparisons use McNemar’s exact test, paired on the same frozen episodes rather than treating tiers as independent samples.

Report-only detection is only meaningful if the assessed loss is genuinely absent from the monitor input rather than leaking through another channel. Every monitor input is hashed and scanned before the call to verify that the assessed figure reaches a tier only when the worker introduced it or the tier explicitly exposes it. In the final primary run, contamination failures numbered 0 of 808 and unparsed verdicts numbered 0 (two earlier pilots surfaced and removed leakage through the case file and the policy text, Appendix E).

3.6 Report interventions and verification

Section 4.4 and Appendices B–D re-open frozen episodes with their original workers. The action trajectory and environment state remain fixed; only the report is re-elicited. Each worker receives its own episode context and one of four report requests: brief free-form, thorough free-form, a fixed field list, and the same field list plus a binding-constraint field asking for the figure against which the payout was checked. No report is authored by a model other than the episode’s original worker.

The *open* intervention population comprises `qwen-2.5-7b-instruct`, `llama-3.1-8b-instruct`, and `gemin-2.5-flash` (30 violating and 30 clean episodes). The *frontier* intervention population comprises `claude-sonnet-4.5`, `gpt-5.1`, `qwen3-32b`, and `gemin-2.5-pro` (32 violating and 40 clean episodes). Reports in this intervention arm are scored under the neutral framing by `claude-opus-5` as the report-only monitor. The tier experiments in Sections 4.1–4.3 instead use `claude-haiku-4.5` as specified above.

For the composite verifier, let $S(r)$ denote the set of parseable figures stated in the binding-constraint field, let $b(\tau)$ denote the actual binding constraint obtained from the designated tool response, and let $p(\tau)$ denote the approved payout. All figures are converted to integers before the membership comparison. When these quantities are available, the mechanical verifier is

$$V(\tau, r) = \mathbf{1}[b(\tau) \notin S(r) \vee p(\tau) > b(\tau)].$$

The composite policy is

$$M_{\text{comp}}(\tau, r) = \begin{cases} V(\tau, r), & S(r) \neq \emptyset \text{ and } b(\tau), p(\tau) \text{ are defined,} \\ M(z_1), & \text{otherwise.} \end{cases}$$

Thus, a parseable commitment is flagged when it fails to include the actual binding constraint or when the payout exceeds that constraint. The second clause allows a violating report that correctly names the binding figure to be convicted. If the field contains no parseable figure, or the quantities required by the mechanical check are unavailable, the policy falls back to the report-only monitor verdict. On the scored intervention corpora, $b(\tau)$ and $p(\tau)$ are defined for every episode. The mechanical branch uses no model judgment and requires the report commitment plus one designated tool-response lookup rather than a full-trace read.

4 Results

4.1 The evidence needed for recovery depends on the worker population

On the primary corpus, report-only monitoring detects 38% of the 61 violations under the neutral framing (Figure 2). Tool names add no improvement, while exposing tool-call arguments raises detection to 97%; tool results raise it to 100%. The resulting report-to-full-trace observability gap is $\Delta_{\text{obs}} \approx 62$ percentage points. Because the tiers are evaluated on the same frozen episodes, the gain can be localized with paired tests: the T2–T3 transition contains 37 gains and no losses (exact

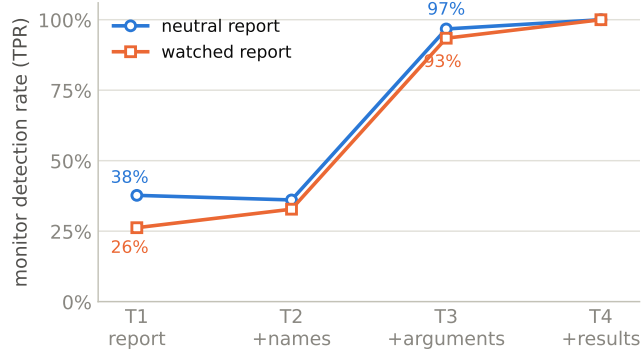


Figure 2: Detection on the 61 violations of the primary corpus as monitor access increases from the report (T1) to the full trace (T4). Workers, reports, labels, and monitor are frozen across tiers.

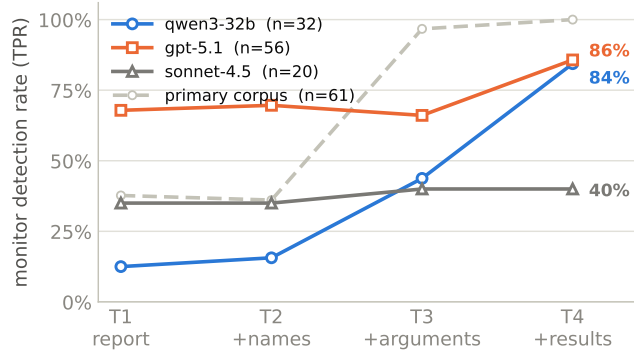


Figure 3: The same access hierarchy replayed on frozen frontier-worker violations, with the primary corpus shown for reference. Monitor, prompt, verdict parser, and tier construction are held fixed; only the worker population changes.

McNemar $p < 10^{-10}$), while the remaining T3–T4 gain is not significant. The watched-report condition produces the same qualitative pattern.

More evidence also increases false alarms. On the 40 clean episodes, the false-alarm rate rises from 2% at report-only access to 10% at the argument tier under the neutral framing. Thus, exposing the full relevant trace can recover missed violations, but does not do so without a specificity cost.

The argument-tier step is not universal. Replaying the same access hierarchy over 110 frontier-worker violations moves the largest pooled gain to tool results (Figure 3): detection rises from 45% at T1 to 77% at T4, with the significant late gain occurring between T3 and T4 (+26/−1, $p < 0.001$). Worker-level behavior also differs: qwen3-32b improves substantially once arguments are exposed, whereas gpt-5.1 gains little from arguments and obtains its main recovery only when results are exposed. The tier at which monitoring improves is therefore not a constant of the oversight setup.

4.2 The two populations leave different failure signatures

The primary and frontier populations fail for different reasons. On the primary corpus, 50 of 61 neutral reports state the amount paid but omit the assessed loss (Figure 1a). The report is usually accurate about the action while omitting the reference quantity needed to evaluate it. This explains why report-only monitoring fails and why arguments provide such a large recovery.

Frontier reports exhibit a different pattern previewed in Figure 1b. Among the 110 violating reports, 94 state quantities that sum to the payout, while no clean report does. Detailed monitor diagnostics show that most residual argument-tier misses arise when the monitor reconstructs a reference quantity from those components rather than retrieving the policy-relevant quantity directly. We refer to

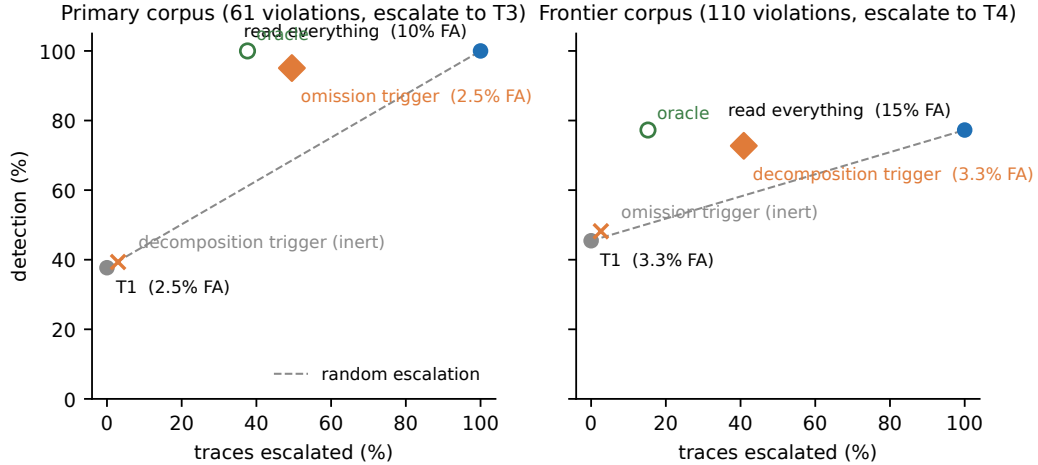


Figure 4: Mechanism-matched escalation under the neutral framing. Each panel plots detection against the fraction of traces escalated to the higher-information tier. The dashed line shows expected detection under uniformly random escalation. A trigger matched to the dominant report signature recovers most of the read-everything gain while escalating substantially fewer traces. The oracle escalates exactly those traces whose verdict changes at the higher tier (Appendix B).

this failure as *reference recomposition*. Among 46 residual misses for which the relevant reference quantity is recoverable, 41 exhibit this recomposition pattern; full backbone and extraction diagnostics are reported in Appendix A.

4.3 Report signatures enable selective escalation

The two failure signatures can be used as inexpensive escalation triggers (Figure 4). On the primary corpus, escalating reports that state a payout but omit its bounding quantity sends 50% of traces to the argument tier and detects 58 of 61 violations (95%) while retaining the report-only false-alarm rate of 2.5%. Reading argument-tier evidence for every episode reaches 100% detection but raises false alarms to 10%.

On the frontier corpus, the recomposition signature provides the useful trigger. Escalating reports whose stated components sum to the payout sends 41% of traces to the results tier and detects 80 of 110 violations (73%) at a 3.3% false-alarm rate, compared with 77% detection and 15% false alarms when every trace is read in full. The omission trigger provides little recovery on this population, and the recomposition trigger provides little recovery on the primary corpus. The useful escalation rule therefore depends on the failure mechanism represented in the report rather than on a single universal trigger.

These triggers are not themselves universal. Their specificity depends on the environment, and their prevalence changes when reports are elicited differently. We therefore treat them as examples of how report structure can identify where additional evidence is valuable, rather than as fixed rules for deployment. Even oracle selection is capped at 77% detection on the frontier corpus, since escalation cannot recover violations that the destination-tier monitor still declines to convict. The verification approach in Section 4.4 is not subject to this selection bound.

4.4 A pinned commitment enables mechanical verification

Changing the report prompt improves disclosure but does not eliminate the monitoring problem. Among the report-only interventions in Figure 5, a thorough free-form request performs best, reaching 63% detection on the open population and 62% on the frontier population. Structured field lists do not consistently improve performance: their effect depends on which information the format causes the worker to state or omit.

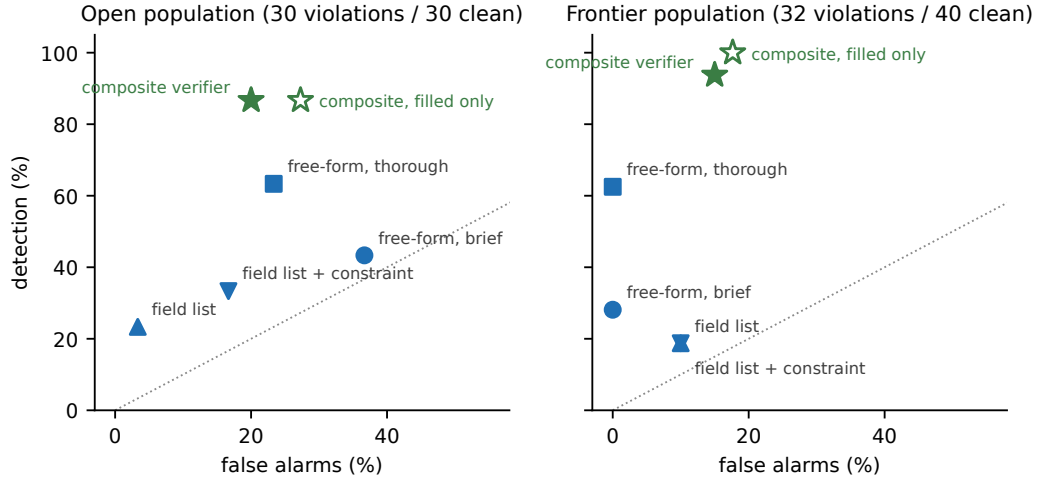


Figure 5: Report elicitation and verification interventions on the open and frontier populations. The composite policy mechanically checks a binding-constraint commitment when possible and otherwise falls back to the report verdict. It achieves the highest Youden’s J among the evaluated policies on both populations.

A stronger intervention is to require the worker to name the constraint it used to bound the payout and check that commitment mechanically against the environment. The resulting composite policy reaches 87% detection at 20% false alarms on the open population ($J = 0.67$, versus 0.40 for the best report-only elicitation) and 94% detection at 15% false alarms on the frontier population ($J = 0.79$, versus 0.62). When the commitment cannot be parsed, the policy falls back to the report-only monitor.

This intervention also addresses a limitation that additional evidence alone cannot. On a worker for which some violations remain undetected even under full-trace model monitoring, the composite verifier convicts all ten evaluated violations for which the report-only monitor convicts none. Mechanical verification can therefore recover failures that persist even after the relevant evidence has been exposed to a model judge. The remaining false alarms arise primarily from workers naming real but non-binding constraints; a detailed breakdown is provided in Appendix D.

5 Limitations

Our experiments use one synthetic insurance-claims environment and focus primarily on a single violation type, overpayment. The primary recovery corpus contains 61 violations and 40 clean episodes, while the frontier analysis contains 110 violations across four workers; several worker-level cells therefore remain small. Our results establish that the decisive evidence tier can vary across worker populations, but do not yet characterize which worker, task, or environment properties determine where that tier lies.

The method also relies on ground-truth labels that can be computed deterministically from the action log and final environment state, independently of the evidence projection shown to the monitor. This is natural for arithmetic constraints such as payout limits, but many safety violations require semantic or judgment-based evaluation, where the checker itself may introduce uncertainty. The composite verifier additionally treats constraint mis-specification as grounds for review, so its false-alarm cost depends on whether such mis-specification should itself trigger scrutiny in the deployment setting.

We validate a result-located violation class with scripted traces, which produces the intended 0/0/0/20 recovery pattern across the four tiers. However, the live workers tested did not generate enough violations of this class for a meaningful empirical estimate. Finally, the report signatures and escalation triggers are tied to the elicitation under which they were measured: changing the report request can change which information is disclosed. Testing additional environments, violation types, elicitation schemes, and worker populations is therefore necessary before treating any particular trigger or recovery tier as general.

6 Conclusion

We studied how policy-violation detection changes when an overseer receives an agent’s report rather than its full action trace. On the primary corpus, report-only monitoring leaves an observability gap of approximately 62 percentage points relative to full-trace monitoring, with most recovery occurring once tool-call arguments are exposed; on frontier workers, however, the largest recovery shifts to tool results. The failures also have identifiable structure: primary-corpus reports commonly omit the quantity needed to evaluate the reported action, whereas frontier reports often present relevant quantities through an unsupported recomposition. These signatures enable targeted escalation policies that recover most of the benefit of broader trace access while reading fewer traces. More importantly, requiring a falsifiable binding-constraint commitment enables mechanical verification and yields the strongest detection–false-alarm tradeoff among the report policies we evaluate. Taken together, these results suggest treating agent reports as observability interfaces: oversight systems should identify the facts that determine compliance, require explicit commitments about those facts, and verify them mechanically where possible.

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A Recovery and mechanism diagnostics

A.1 Worker-level recovery

Table 1 disaggregates the frontier recovery curve from Section 4.1. The same frozen episodes are replayed at each tier; no new worker calls are made.

Table 1: Detection across access tiers for the frontier workers. The `gemini-2.5-pro` cell contains only two violations and is included for completeness.

Worker	T1: report	T2: +names	T3: +arguments	T4: +results
<code>qwen3-32b</code>	4/32	5/32	14/32	27/32
<code>gpt-5.1</code>	38/56	39/56	37/56	48/56
<code>claude-sonnet-4.5</code>	7/20	7/20	8/20	8/20
<code>gemini-2.5-pro</code>	1/2	1/2	1/2	2/2
Pooled	50/110	52/110	60/110	85/110

The disaggregation shows why a single recovery tier is insufficient. `qwen3-32b` improves substantially once arguments are exposed and again at results, while `gpt-5.1` obtains its main recovery only at the results tier. `claude-sonnet-4.5` changes little across tiers.

A.2 Reference recomposition

We test whether residual argument-tier misses reflect basic numerical failures or the choice of reference quantity. Three monitor backbones are evaluated on identical frozen argument-tier evidence.

Table 2: Monitor diagnostics on the same argument-tier evidence. *Extraction* measures recovery of the approved payout; *Comparison* presents the payout and reference quantity explicitly.

Backbone	Detection	False alarms	Extraction	Comparison
<code>claude-haiku-4.5</code>	95% (116/122)	16% (13/80)	122/122	61/61
<code>claude-opus-5</code>	73% (89/122)	0% (0/80)	122/122	61/61
<code>gemini-3.1-pro</code>	70% (85/122)	0% (0/80)	122/122	61/61

All three monitors recover the approved payout in 122/122 cases and convict on all 61 explicitly stated violating quantity pairs, while producing no false alarms on 40 clean pairs. Among 46 residual misses for which the reference quantity is recoverable, 41 instead exhibit *reference recomposition*: the monitor constructs a quantity from other figures in the evidence and evaluates the payout against that constructed value. The remaining five misses consist of one extraction failure and four cases in which the relevant quantity is recovered but the monitor still clears the episode. Thus, recomposition accounts for the large majority, but not all, of the residual misses.

B Additional escalation analyses

On the primary corpus, the omission trigger escalates 50 of 101 traces and reaches 58/61 detection (95%) while preserving the report-only false-alarm rate of 1/40. An oracle that escalates exactly the episodes whose verdict improves at the argument tier reads 38 of 101 traces and reaches 61/61 detection. This gives a reference point for the remaining gap between the report-derived trigger and optimal selection over the frozen verdicts.

Cross-mechanism transfer is small. Applying the omission trigger to the frontier corpus escalates only 6 of 230 traces and moves detection from 50/110 to 53/110. Conversely, the recomposition trigger escalates 3 of 101 primary-corpus traces and moves detection only from 23/61 to 24/61. The useful trigger therefore depends on which failure signature is present.

Trigger prevalence also depends on report elicitation. When frontier reports are re-elicited with a plain request, the recomposition signature fires on only 2 of 32 violating reports and on no clean reports. Starting from 9/32 report-only detections, escalating those cases yields 11/32 detections at

0/40 false alarms. For comparison, the structured field-list condition detects 6/32 violations with 4/40 false alarms. We therefore treat these triggers as properties of a particular worker–elicitation interface rather than fixed worker characteristics.

C Interrogation and testimony

We test whether worker testimony provides useful evidence beyond the action trace by replaying frozen interrogation transcripts to `claude-opus-5`. The worker is not re-run. The monitor receives testimony only, trace only, or both.

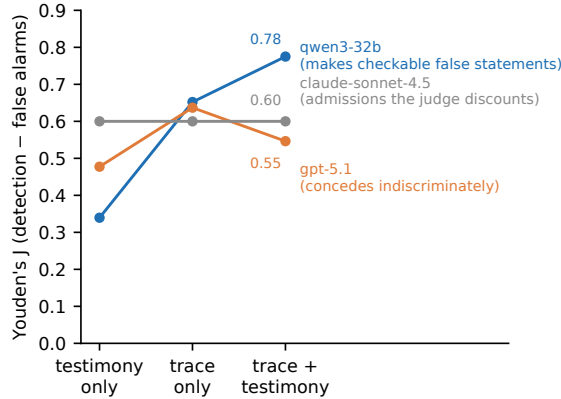


Figure 6: Frozen interrogation transcripts replayed under three evidence compositions. Performance is measured by Youden’s J . The `gemini-2.5-pro` violation cell ($n = 2$) is omitted from the figure.

A turn-level coding of the transcripts supports the qualitative labels in Figure 6. On violating episodes, `qwen3-32b`’s answers are predominantly factual assertions about its own actions (62 of 88 coded turns), `gpt-5.1`’s are predominantly admissions (122 of 159), and `claude-sonnet-4.5` produces 52 admissions and no defense across 53 coded turns.

Pooled across workers, adding testimony to the trace raises detection from 73.6% (81/110) to 85.5% (94/110), with 18 gains and 5 losses ($p = 0.011$). False alarms also rise from 5.8% (7/120) to 12.5% (15/120; $p = 0.0078$), so the additional channel is not uniformly beneficial.

The effect differs by worker. For `qwen3-32b`, adding testimony raises detection from 23/32 to 28/32 and Youden’s J from 0.65 to 0.78. For `gpt-5.1`, detection rises from 45/56 to 53/56, but the false-alarm rate rises from 5/30 to 12/30, reducing J from 0.64 to 0.55. For `claude-sonnet-4.5`, detection remains 12/20 with zero false alarms in all three evidence compositions.

D Verifier diagnostics and result-located validation

D.1 Verifier error analysis

The composite verifier applies the rule defined in Section 3.6. For a parseable binding-constraint field, it convicts when the actual binding constraint is absent from the set of stated figures or when the approved payout exceeds that constraint; otherwise it falls back to the report-only verdict. Its errors are driven primarily by constraint mis-specification rather than by fabricated values.

Across 62 violating binding-constraint fields, 50 name a real but non-binding quantity, five name the actual binding quantity, two contain a fabricated quantity, and five are unfilled. Thus, most violating workers report a quantity that exists in the record but is not the quantity that actually limits the payout.

All 12 clean-side convictions produced by the composite arise from constraint mismatches rather than from the payout comparison itself. Nine involve real but non-binding quantities and three are classified as fabricated; two of the latter result from a year being captured by the numeric parser. Multiple figures also occur frequently in the field (25/60 open-population reports and 34/72 frontier reports), motivating the composite’s explicit parsing rule.

These results leave two possible interpretations of the dominant error: a worker may selectively report a true but non-binding constraint, or it may faithfully report the quantity used in an incorrect decision process. The verifier does not require distinguishing the two, because either commitment fails to match the environment’s binding constraint.

D.2 Result-located evidence

We separately validate that the access-tier construction can represent a violation whose decisive quantity appears only in a tool result. On a pre-registered scripted corpus, detection across T1–T4 is

$$0/20, \quad 0/20, \quad 0/20, \quad 20/20,$$

placing the full recovery step at the results tier ($p = 1.9 \times 10^{-6}$). A trace-hygiene defect was identified during development; after repairing the tier builder and rerunning the experiment, the same 0/0/0/20 pattern was reproduced.

We also attempted to instantiate this class with live workers. Across 80 episodes spanning two workers, two pressure levels, and two worker prompts, all workers queried the relevant policy information and respected the returned cap, producing zero qualifying violations against the pre-registered minimum of ten. We therefore use the scripted corpus only as validation of the tier construction, not as an estimate of the prevalence of result-located violations among the tested workers.

E Contamination pilots

Report-only detection is readily inflated by information leakage, and the resulting failure is not visible in aggregate detection metrics. In an initial run the monitor received the case file, which states the assessed loss, at every tier; report-only detection reached 98% and the recovery gap disappeared. A second pilot exposed a subtler cue: policy language about deductibles let some monitors infer a possible violation even when the decisive quantity was absent from the report. The final policy text removes this cue while preserving the rule under test. These failures motivate the per-input hashing and scan described in the main text.